



Systems Engineer (Firmware)

Location

Menlo Park, CA

Employment Type

Full time

Location Type

On-site

Department

Software ›

Platform

OverviewApplication

Company Overview

Each year, 2.5 trillion hours are spent on household chores. At Matic, we're on a mission to recapture that lost time, and we're doing it by revolutionizing home robotics.

Our first product, also called Matic, is a Wall-E-esque floor cleaning robot. We've built what amounts to "full self-driving in the home" with real-time 3D mapping, adaptive path-tracking, and a precise semantic understanding of the home. Our breakthroughs in spatial AI allow Matic to work reliably in real homes, using only RGB cameras and neural networks running on-device.

Privacy First

What happens in the home, stays in the home. Our robots are private by design, with all data processing performed by the robot itself, not in the cloud.

Our Approach

Before the iPhone, consumers adopted several distinct devices; cell phones, PDAs, and portable music players each served a particular need. We believe in a similar progression for home robotics, starting with single-purpose robots and building iteratively toward more complex capabilities over time.

Our Culture

Matic is a tight-knit and collaborative team, singularly focused on building products our customers will cherish. We're ultra-hardworking people committed to solving tough problems that save precious time and energy.

About the role

As the name "firmware" may imply, this role closes the gap between our software and our hardware. In this role, you will be the hardware team's window into Matic's software, and you will be the software team's window into Matic's hardware. Firmware engineers are fundamentally cross-discipline and are expected to operate with product-focused systems-level thinking.

What you'll do

Firmware at Matic is a broad domain, and you will work on a wide variety of projects such as:

- Bring-up and support for hardware changes (electrical and mechanical). For example:
 - A new sensing mechanism to detect when the waste bag is full from water during mopping - requires updating device drivers to interface with the new sensor, implementing tests to ensure the new system is working as expected during assembly, and coordinating robot behaviors to accommodate the new system (data filtering, updating the bag fullness check).
- Design and improve the behavioral logic and controls algorithms for Matic's physical systems
- Analyze robot logs along with motor and sensor data to debug hardware failures. For example:
 - Customer reports their Matic isn't charging despite being properly docked
 - Employee testing their Matic at home mentions that their robot suddenly started making weird clicking sounds while mopping

- Support internal processes with tooling to interface with Matic's motors and sensors

What we look for

We're looking for strong software experience with electromechanical projects. This likely includes deep experience with microcontrollers (bare-metal programming, RTOS, communication protocols), a fundamental understanding of electronics (schematics, datasheets, motors, sensors, batteries), and intuition for physical systems (motors, moving parts, materials). You should have strong experience writing embedded C/C++, and we hope you are eager to learn Rust if you aren't already familiar with it (see: [why we use Rust](#)).

Must haves:

- 2+ years of industry experience or equivalent (e.g. grad school)
- Core software skills:
 - data structures, control flow, computer systems
 - deep experience with systems programming language(s): C, C++, rust, etc.
 - experience / comfort with scripting / prototyping language(s): python, etc.
- Experience with hardware (the more the better):
 - Motors (brushed and brushless DC, drivers, operating principles)
 - Controls systems
 - Sensors (datasheets, device drivers, data filtering)
 - Electrical fundamentals
 - Mechanical intuition
 - You've built things (robots, PCBs, 3D-printing, anything)
- Experience with microcontrollers (STM32, PIC, ESP32, etc.)
- Soft-skills:
 - Multi-disciplinary team & leadership experience
 - Thinks broadly, connecting insights from across the system to solve problems
 - Meticulous, detail oriented, organized
 - Fixated on edge-cases and testing

Nice to have:

- Friendly & approachable personality
 - non-combative, selfless, team-first mentality
- Higher-level robotics exposure:
 - computer vision, machine learning, path planning, SLAM

We'd love to hear from you if...

- You love diagnosing complex electromechanical issues
- You are genuinely motivated to help those around you
- You are passionate about learning outside of your normal comfort zones
- You are excited to do great work

Compensation

Base \$120,000 — \$200,000

Actual Compensation will depend on skills, experience and qualifications. Base Salary is one part of the total compensation package. The role is also eligible for equity through the company's discretionary equity program, along with a comprehensive benefits package that includes medical, dental and vision coverage access to a 401(k) plan.

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